

RESPUESTAS TALLER 8 NÚMEROS COMPLEJOS

1.

$$a) (-1 + 4i) \quad b) (2 - 13i) \quad c) (-9 - 15i) \quad d) 4 \quad e) (-198 - 10i) \quad f) (6 - 4i) \quad g) i$$

$$h) \left(\frac{-6}{53} + \frac{32}{53}i \right) \quad i) \left(\frac{11}{2} + \frac{9}{2}i \right) \quad j) \left(\frac{-6}{13} - \frac{4}{13}i \right) \quad k) (9 + i) \quad l) \left(\frac{-8}{5} - \frac{26}{5}i \right)$$

2.

$$a) \left(\frac{-116}{17} + \frac{29}{17}i \right) \quad b) \left(\frac{-17}{8}i \right) \quad c) \left(\frac{-400}{29} - \frac{160}{29}i \right)$$

3.

$$a) x = 2; y = \frac{3}{2} \quad b) x = -9; y = -20 \quad c) x = \frac{1}{5}; y = -\frac{2}{5}$$

4.

$$x_1 = \left(\frac{-1}{4} + \frac{\sqrt{15}}{4}i \right) \quad x_2 = \left(\frac{-1}{4} - \frac{\sqrt{15}}{4}i \right)$$

5.

$$x_1 = -\frac{1}{3} - \frac{\sqrt{14}}{3}i \quad x_2 = -\frac{1}{3} + \frac{\sqrt{14}}{3}i$$

6.

$$x_1 = \frac{1}{8} + \frac{\sqrt{31}}{8}i \quad x_2 = \frac{1}{8} - \frac{\sqrt{31}}{8}i$$

7.

$$a) z = \frac{6}{13} - \frac{4}{13}i \quad b) z = \frac{3}{17} + \frac{12}{17}i$$

8.

$$a) x = -\frac{4}{9} \quad b) x = \frac{1}{4}$$

9.

$$x_1 = 2 \quad x_2 = -1$$

10.

$$a) |z| = 1; \angle = 300^\circ \quad b) |z| = 5; \angle = 36^\circ \quad c) |z| = 3\sqrt{2}; \angle = 290^\circ \quad d) |z| = \sqrt{29}; \angle = 158^\circ$$

$$e) |z| = 2\sqrt{17}; \angle = 194^\circ \quad f) |z| = \sqrt{5}; \angle = 334^\circ \quad g) |z| = \sqrt{2}; \angle = 225^\circ$$



11.

$$\begin{aligned} & a) 4 \left(\cos \frac{3\pi}{2} + i \operatorname{sen} \frac{3\pi}{2} \right) \quad b) 15 \left(\cos \frac{\pi}{2} + i \operatorname{sen} \frac{\pi}{2} \right) \quad c) 10 \left(\cos \frac{\pi}{6} + i \operatorname{sen} \frac{\pi}{6} \right) \\ & d) \sqrt{29} \left(\cos \frac{28\pi}{45} + i \operatorname{sen} \frac{28\pi}{45} \right) \quad e) \sqrt{34} \left(\cos \frac{301\pi}{180} + i \operatorname{sen} \frac{301\pi}{180} \right) \quad f) 2\sqrt{2} \left(\cos \frac{5\pi}{4} + i \operatorname{sen} \frac{5\pi}{4} \right) \\ & g) \sqrt{2} \left(\cos \frac{7\pi}{4} + i \operatorname{sen} \frac{7\pi}{4} \right) \end{aligned}$$

12.

$$a) -6 + 0i \quad b) -16\sqrt{3} + 16i \quad c) -4\sqrt{3} - \frac{1}{2}i \quad d) \frac{-\sqrt{3}}{4} - \frac{1}{4}i$$

13.

$$a) -1 + 0i \quad b) 2\sqrt{2} + 2\sqrt{2}i \quad c) -\frac{243\sqrt{3}}{2} - \frac{243}{2}i \quad d) 0 - 16i$$

14.

$$\begin{aligned} & a) -1 + 0i \quad b) 0 - 8i \quad c) -64 + 0i \quad d) -16\sqrt{3} + 16i \quad e) 0 + (\sqrt{2})^9 i \\ & f) 128 - 128i \quad g) 512 + 512\sqrt{3}i \end{aligned}$$

15.

$$\begin{aligned} & a) W_0 = 1 + \sqrt{3}i \quad W_1 = -2 + 0i \quad W_3 = 1 - \sqrt{3}i \\ & b) W_0 = 0,92 + 0,38i \quad W_1 = -0,38 + 0,92i \quad W_2 = -0,92 - 0,38i \quad W_3 = 0,38 - 0,92i \\ & c) W_0 = 0,59 + 1,02i \quad W_1 = -1,02 + 0,59i \quad W_2 = -0,59 - 1,02i \quad W_3 = 1,02 - 0,59i \\ & d) W_0 = 1,09 + 0,45i \quad W_1 = -1,09 - 0,45i \\ & e) W_0 = 1,96 + 0,3i \quad W_1 = 0,7 + 1,86i \quad W_2 = -1,24 + 1,54i \quad W_3 = -1,96 - 0,3i \\ & \quad W_4 = -0,7 - 1,86i \quad W_5 = 1,24 - 1,54i \end{aligned}$$

16.

$$a) W_0 = 3 + 0i \quad W_1 = 0 + 3i \quad W_2 = -3 + 0i \quad W_3 = 0 - 3i$$

$$b) W_0 = 2 + 0i \quad W_1 = -1 + \sqrt{3}i \quad W_2 = -1 - \sqrt{3}i$$

$$c) W_0 = \sqrt{3} + i \quad W_1 = -\sqrt{3} + i \quad W_2 = 0 - 2i$$

$$d) W_0 = \frac{3\sqrt{3}}{2} + \frac{3}{2}i \quad W_1 = -\frac{3\sqrt{3}}{2} + \frac{3}{2}i \quad W_2 = 0 - 3i$$

$$e) W_0 = 1,14 + 0,11i \quad W_1 = 0,24 + 1,12i \quad W_2 = -0,99 + 0,57i \quad W_3 = -0,85 - 0,77i$$

$$W_4 = 0,47 - 1,05i$$

$$f) z_1 = \sqrt{\frac{-2 + \sqrt{5}}{2}} + \frac{1}{\sqrt{2}\sqrt{\sqrt{5} - 2}}i \quad z_2 = -\sqrt{\frac{-2 + \sqrt{5}}{2}} - \frac{1}{\sqrt{2}\sqrt{\sqrt{5} - 2}}i$$

$$z_3 = \sqrt{\frac{1 + \sqrt{5}}{2}} - \sqrt{\frac{\sqrt{5} - 1}{2}}i \quad z_4 = -\sqrt{\frac{1 + \sqrt{5}}{2}} + \sqrt{\frac{\sqrt{5} - 1}{2}}i$$